Wednesday Warm Up: Unit 4, Inductance, Reactance, RL circuits, RLC circuits

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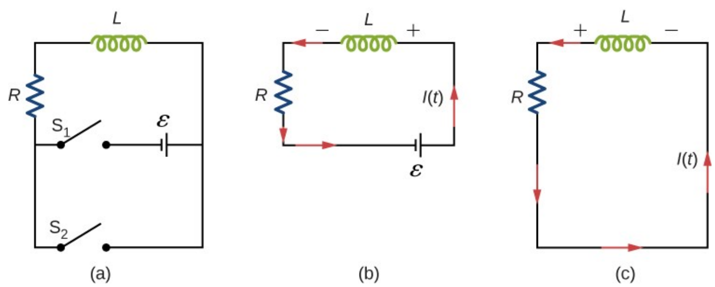

Figure 1: A circuit diagram of a DC emf with a resistor, inductor, and two switches.

1 Memory Bank

- $\epsilon = -L\frac{\Delta I}{\Delta t}$ ... Faraday's law with induction, L. Think of this as the change in voltage across an inductor.
- $X_{\rm C} = 1/(2\pi f C)$ ... The reactance of a capacitor with capacitance C, at frequency f.
- $X_{\rm L} = 2\pi f L$ ... The reactance of an inductor with inductance L, at frequency f.
- $Z_{\rm tot} = \sqrt{R^2 + (X_{\rm L} - X_{\rm C})^2}$ ... The total impedance of an RLC series circuit.
- $\tau = RC$ ... The time constant of an RC circuit.
- $f_0 = \frac{1}{2\pi\sqrt{LC}}$ ... RLC circuit resonance frequency.
- $i(t) = \frac{\epsilon}{R}\left(1 - \exp(-t/\tau)\right)$ ... Current in a charging RL circuit.
- $i(t) = \frac{\epsilon}{R}\exp(-t/\tau)$ ... Current in a discharging RL circuit.
- $\tau = L/R$ ... Time constant of an RL circuit.
- $V_{\rm rms} = I_{\rm rms} Z_{\rm tot}$ ... Ohm's Law for AC circuits.

2 RL Circuits

1. Consider Fig. 1. There is a battery connected to a resistor R and and inductor L via switch 1. Switch 2 connects L and R. (a) What is the current at $t = 0$, when switch 1 is closed? (b) What is the current when $t \gg \tau$? (This is like part (b) of Fig. 1).
2. (a) What is τ when $R = 1$ kΩ, and $L = 0.2$ Henries? (b) What will the time constant τ be if we double R and quadruple L?
3. If switch 1 is open and switch 2 closed after $t \gg \tau$, what time passes before the current is reduced to half of its original value?

3 Reactance and Impedance

1. (a) What is the reactance, in ohms, of a 2.5 μF capacitor at 1400 kHz? (b) What is the reactance of a 5.0 mH inductor at this same frequency? (c) Now add (in series) a 1 kΩ resistor. What is the total impedance at at this frequency? (d) What is the total impedance at 1.4 MHz?
2. What is the resonance frequency of the RLC circuit in the previous exercise?